Hilbert Basis Theorem and Affine Finiteness

An End-to-End Meta-Math Walkthrough

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(Dated: October 2026)

Abstract

Collect possible coefficients at each degree into an ascending chain of ideals. Once that chain stabilizes, finitely many chosen polynomials reduce every other polynomial by decreasing degree. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AG-01 / Algebraic Geometry / Report v1

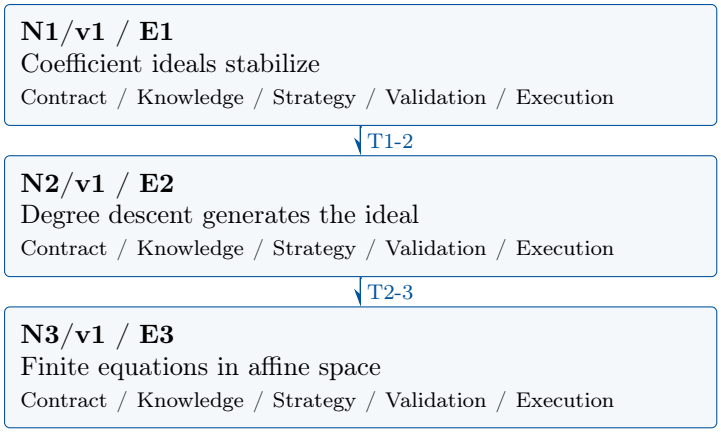
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *If R is a commutative Noetherian ring with identity (every ideal is finitely generated), then $R[x]$ is Noetherian. Deduce by induction that every ideal of $k[x_1, \ldots, x_n]$ is finitely generated for a field k, so every affine algebraic set is cut out by finitely many polynomial equations.*

A. Foundations and permitted imports

Use elementary ideal arithmetic and polynomial coefficients, including degree reduction by cancellation of a leading coefficient. Noetherianity implies the ascending chain condition: the union of an ascending chain is an ideal; its finitely many generators all lie in one term, which therefore equals the union. For an ideal I, $V(I)$ denotes its common zero set over the chosen field. No algebraic closure hypothesis is needed here.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. There is N with $A_d = A_N$ for all $d \geq N$, and each A_d for $d \leq N$ has finitely many chosen polynomial lifts in I.

Receiving activity. N2.1

Compatibility. E1/v1 and the ideal J generated in $R[x]$ by the finite chosen family. Check: The degree must strictly decrease after cancellation, even when R has zero divisors.

T2-3: N2/E2/v1 to N3. $I = J$, so $R[x]$ is Noetherian.

Receiving activity. N3.1

Compatibility. E2/v1 and a field k, whose only ideals are 0 and k. Check: Equality of zero sets follows from linear combinations of equations, not from radical-ideal correspondence.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Coefficient ideals stabilize

Object and role. Encode infinitely many polynomial constraints in an ascending chain inside the Noetherian coefficient ring. **Local obligation:** There is N with $A_d = A_N$ for all $d \geq N$, and each A_d for $d \leq N$ has finitely many chosen polynomial lifts in I.

Inputs. An ideal $I \subset R[x]$ and $A_d = \{\text{coefficient of } x^d \text{ in } f : f \in I, \deg f \leq d\}$, including zero.

C — Contract

Publish E1/v1 only after establishing this obligation: There is N with $A_d = A_N$ for all $d \geq N$, and each A_d for $d \leq N$ has finitely many chosen polynomial lifts in I.

K — Knowledge

An ideal $I \subset R[x]$ and $A_d = \{\text{coefficient of } x^d \text{ in } f : f \in I, \deg f \leq d\}$, including zero. Retain the stated assumptions and import versions.

S — Strategy

Encode infinitely many polynomial constraints in an ascending chain inside the Noetherian coefficient ring.

V — Validation

Use coefficient ideals at degree at most d; a set of leading coefficients at exactly one degree requires care under cancellation.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Addition and scalar multiplication of polynomials of degree at most d show that A_d is an ideal of R. Multiplication by x shows $A_d \subset A_{d+1}$. The ascending chain condition gives stabilization at some N.

N1.2 For $0 \leq d \leq N$, choose generators $a_{d,1}, \ldots, a_{d,r_d}$ of A_d. Omit zero generators. Choose $f_{d,j} \in I$ of degree at most d whose coefficient at x^d is $a_{d,j}$. A nonzero chosen coefficient makes its degree exactly d. There are only finitely many chosen polynomials.

Export E1/v1. There is N with $A_d = A_N$ for all $d \geq N$, and each A_d for $d \leq N$ has finitely many chosen polynomial lifts in I.

Acceptance gate. Use coefficient ideals at degree at most d; a set of leading coefficients at exactly one degree requires care under cancellation.

Targeted repair. If a sum cancels leading coefficients, retain the coefficient-at-degree definition, which includes zero and remains an ideal. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Degree descent generates the ideal

Object and role. Cancel the highest coefficient of an arbitrary polynomial in I and use induction on its degree. **Local obligation:** $I = J$, so $R[x]$ is Noetherian.

Inputs. E1/v1 and the ideal J generated in $R[x]$ by the finite chosen family. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: $I = J$, so $R[x]$ is Noetherian.

K — Knowledge

E1/v1 and the ideal J generated in $R[x]$ by the finite chosen family. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Cancel the highest coefficient of an arbitrary polynomial in I and use induction on its degree.

V — Validation

The degree must strictly decrease after cancellation, even when R has zero divisors.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Certainly $J \subset I$. Let nonzero $f \in I$ have degree m and leading coefficient a. Set $d = \min(m, N)$. Stabilization gives $a \in A_m = A_d$ when $m > N$, and membership is immediate when $m \leq N$. Write $a = \sum_j b_j a_{d,j}$ with $b_j \in R$.

N2.2 The polynomial $h = f - \sum_j b_j x^{m-d} f_{d,j}$ lies in I and has degree less than m or is zero. Strong induction on the nonnegative integer m proves $h \in J$, hence $f \in J$. At $m = 0$, cancellation leaves zero, which starts the induction. No division by a leading coefficient was used.

Export E2/v1. $I = J$, so $R[x]$ is Noetherian.

Acceptance gate. The degree must strictly decrease after cancellation, even when R has zero divisors.

Targeted repair. If polynomial long division over a field is used for a general ring, replace it by the generator combination above; invertibility is not assumed. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Finite equations in affine space

Object and role. Iterate the one-variable theorem and translate ideal generation into equality of zero sets. **Local obligation:** Every $I \subset k[x_1, \ldots, x_n]$ has finite generators $g_1, \ldots, g_s$ and $V(I) = V(g_1, \ldots, g_s)$.

Inputs. E2/v1 and a field k, whose only ideals are 0 and k. Incoming proof exports: E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: Every $I \subset k[x_1, \ldots, x_n]$ has finite generators $g_1, \ldots, g_s$ and $V(I) = V(g_1, \ldots, g_s)$.

K — Knowledge

E2/v1 and a field k, whose only ideals are 0 and k. Incoming proof exports: E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Iterate the one-variable theorem and translate ideal generation into equality of zero sets.

V — Validation

Equality of zero sets follows from linear combinations of equations, not from radical-ideal correspondence.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 A field is Noetherian. Apply E2 successively to k, $k[x_1]$, and so on to obtain Noetherianity of $k[x_1, \ldots, x_n]$. The induction length is the finite number of variables.

N3.2 If $I = (g_1, \ldots, g_s)$, a point annihilating all generators annihilates each finite polynomial combination of them, hence all of I. The reverse implication is immediate because the generators belong to I. This proves the finite-equation conclusion.

Export E3/v1. Every $I \subset k[x_1, \ldots, x_n]$ has finite generators $g_1, \ldots, g_s$ and $V(I) = V(g_1, \ldots, g_s)$.

Acceptance gate. Equality of zero sets follows from linear combinations of equations, not from radical-ideal correspondence.

Targeted repair. If finite generation is claimed for infinitely many variables, restore the finite-variable contract before reusing the induction. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Coefficient ideals stabilize

1. **N1.1.** Addition and scalar multiplication of polynomials of degree at most d show that A_d is an ideal of R. Multiplication by x shows $A_d \subset A_{d+1}$. The ascending chain condition gives stabilization at some N.

2. **N1.2.** For $0 \leq d \leq N$, choose generators $a_{d,1}, \ldots, a_{d,r_d}$ of A_d. Omit zero generators. Choose $f_{d,j} \in I$ of degree at most d whose coefficient at x^d is $a_{d,j}$. A nonzero chosen coefficient makes its degree exactly d. There are only finitely many chosen polynomials.

B. N2 — Degree descent generates the ideal

1. **N2.1.** Certainly $J \subset I$. Let nonzero $f \in I$ have degree m and leading coefficient a. Set $d = \min(m, N)$. Stabilization gives $a \in A_m = A_d$ when $m > N$, and membership is immediate when $m \leq N$. Write $a = \sum_j b_j a_{d,j}$ with $b_j \in R$.

2. **N2.2.** The polynomial $h = f - \sum_j b_j x^{m-d} f_{d,j}$ lies in I and has degree less than m or is zero. Strong induction on the nonnegative integer m proves $h \in J$, hence $f \in J$. At $m = 0$, cancellation leaves zero, which starts the induction. No division by a leading coefficient was used.

C. N3 — Finite equations in affine space

1. **N3.1.** A field is Noetherian. Apply E2 successively to k, $k[x_1]$, and so on to obtain Noetherianity of $k[x_1, \ldots, x_n]$. The induction length is the finite number of variables.

2. **N3.2.** If $I = (g_1, \ldots, g_s)$, a point annihilating all generators annihilates each finite polynomial combination of them, hence all of I. The reverse implication is immediate because the generators belong to I. This proves the finite-equation conclusion.

□

V. WORKED ORCHESTRATION

In $k[x, y]$, let I be generated by $x^2, xy, xy^2, xy^3, \ldots$. Since $xy^m = y^{m-1}(xy)$ for $m \geq 1$, $I = (x^2, xy)$. Its common zeros satisfy $x = 0$, hence $V(I)$ is the y-axis. Finite generation does not say I is radical: x vanishes on that axis but $x \notin I$.

A. Boundary and hypothesis audit

The ring $k[x_1, x_2, \ldots]$ is not Noetherian. If the ideal $(x_1, x_2, \ldots)$ had finitely many generators, they would involve only finitely many variables. Setting those variables to zero would annihilate every proposed generator but leave a later variable nonzero, contradicting generation.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 activity: divide by the leading coefficient of a polynomial over R. In $\mathbb{Z}[x]$, the coefficient 2 need not be invertible. **Diagnosis:** Strategy substituted a field division algorithm for a ring argument. **Repair:** express the leading coefficient in the ideal generated by the $a_{d,j}$ and subtract the corresponding polynomial combination. The degree falls without division; recheck E2 and the affine consequence. The theorem remains unchanged.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use finite polynomial descriptions as input to elimination and local computations; finite generation alone gives no practical degree bound.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. J. S. Milne, Algebraic Geometry, chapters on commutative algebra, algebraic sets, and affine varieties; author's mathematics site.